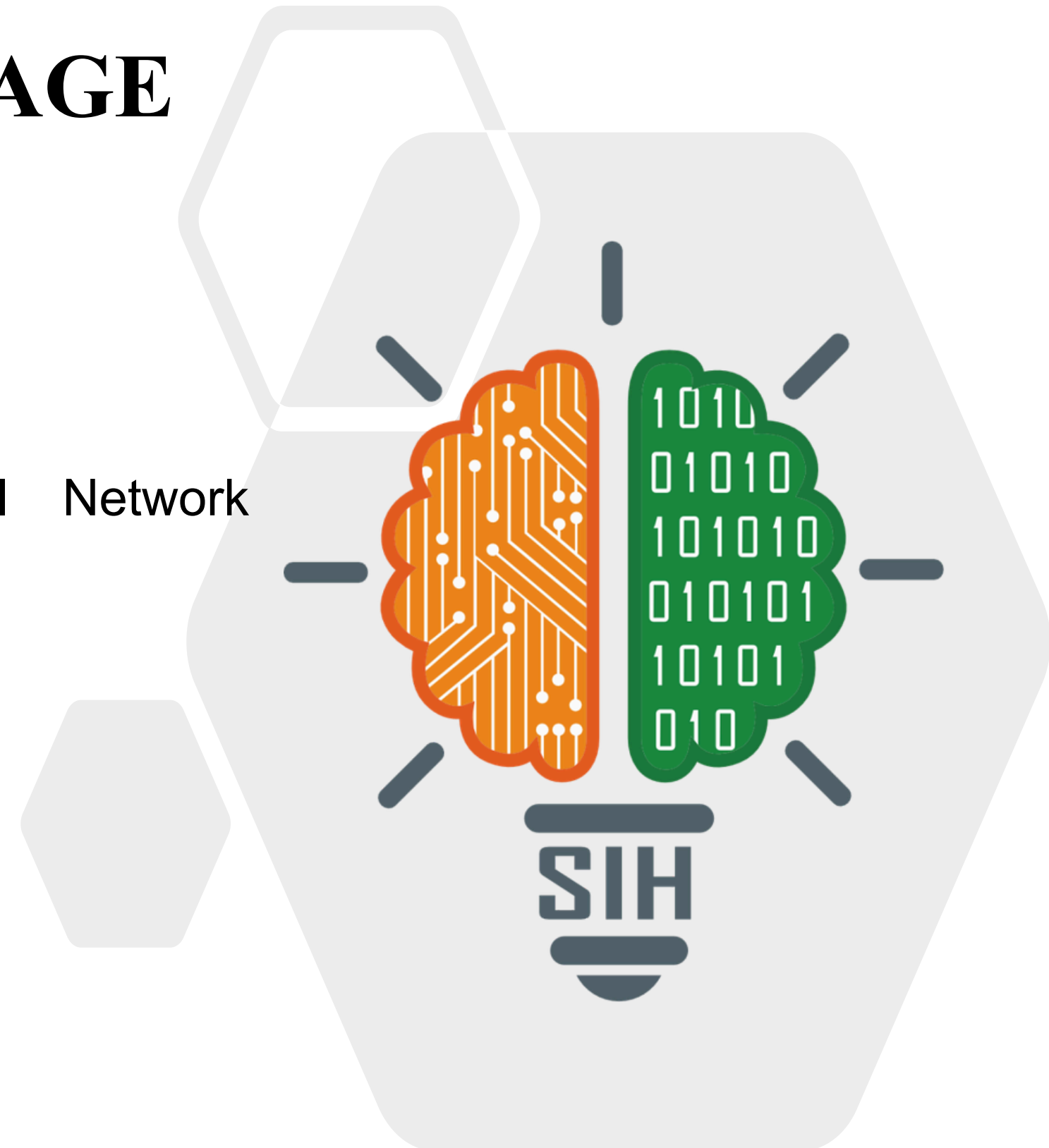




TITLE PAGE

- **Problem Statement ID – SIH26189**
- **Problem Statement Title-** AI-Powered Criminal Network Analysis System .
- **Theme - Blockchain & Cybersecurity**
- **PS Category - Software**
- **Team ID -**
- **Team Name :- Team Venom**



IDEA TITLE

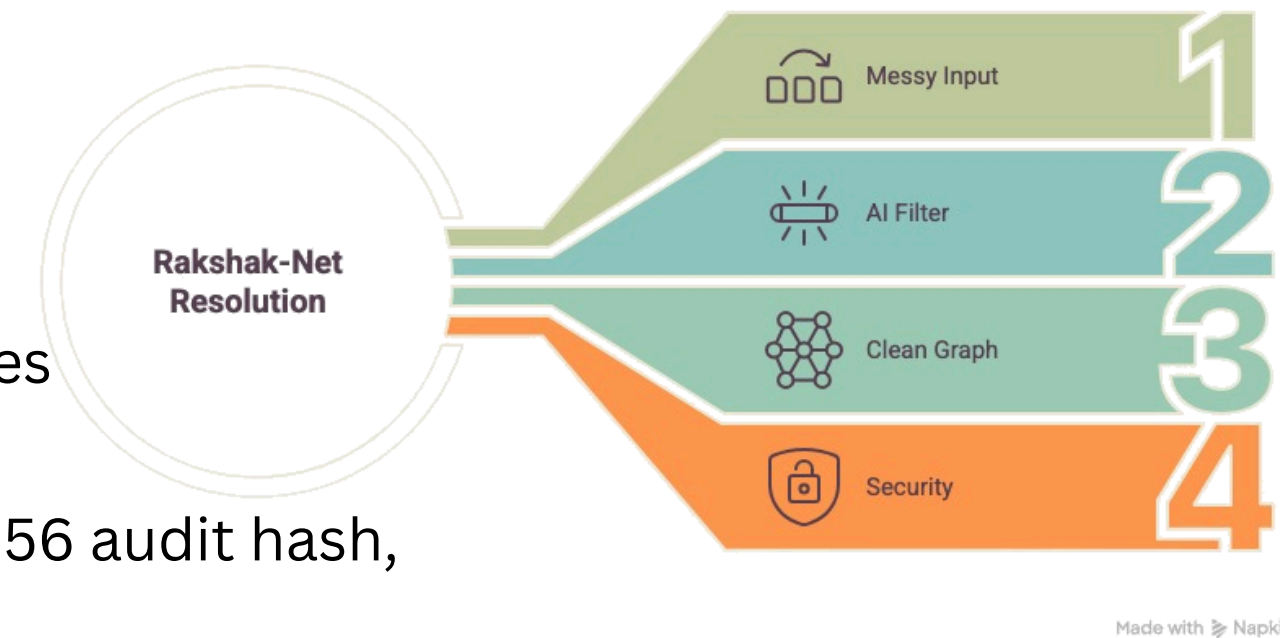
Rakshak AI — Sovereign Police Intelligence & Self-Securing Infrastructure

Rakshak AI is an on-premise criminal network analysis platform that transforms fragmented police data into connected, searchable and evidence-grounded intelligence, while continuously securing its own infrastructure.

How it addresses the problem :-

- Fuses fragmented FIRs, CDRs, financial trails and surveillance data into a unified canonical schema without immediate legacy migration.
- Resolves identities across Hindi, English and Hinglish, handling name, spelling and transliteration variations such as Mohd ↔ मोहम्मद.
- Builds a dynamic 6-layer Temporal Knowledge Graph connecting Persons ↔ Vehicles ↔ Locations ↔ Incidents ↔ Communication ↔ Financial links.
- Grounds every inferred relationship to its source FIR/CDR/document with a SHA-256 audit hash, enabling transparent evidence provenance.
- Supports NCRB Women Safety investigations by mapping repeat-offender networks, intermediaries and hidden connections across cases.
- Runs 100% offline / air-gapped with local AI, while RakshakAI (14B) continuously scans its own infrastructure for security vulnerabilities.

Unveiling Rakshak-Net's Resolution Process



Innovation & Uniqueness :-

Rakshak AI combines four capabilities in one sovereign investigation platform:

FRAGMENTED DATA → ENTITY RESOLUTION → TEMPORAL CRIMINAL GRAPH → EVIDENCE-GROUNDED INTELLIGENCE

Its differentiator is RakshakAI: an offline security layer that protects the same AI infrastructure used for investigation.

SOFTWARE & DATA

Languages & Frameworks

- Backend: Python • FastAPI • Pydantic
- Frontend: Next.js • TypeScript • Tailwind CSS
- Graph Query: Cypher • Neo4j

Databases

- Graph: Neo4j — Multi-layer Temporal Knowledge Graph
- Relational / Vector: PostgreSQL • pgvector

Infrastructure

- Docker
- GitHub Actions — CI/CD

AI / ML & HARDWARE

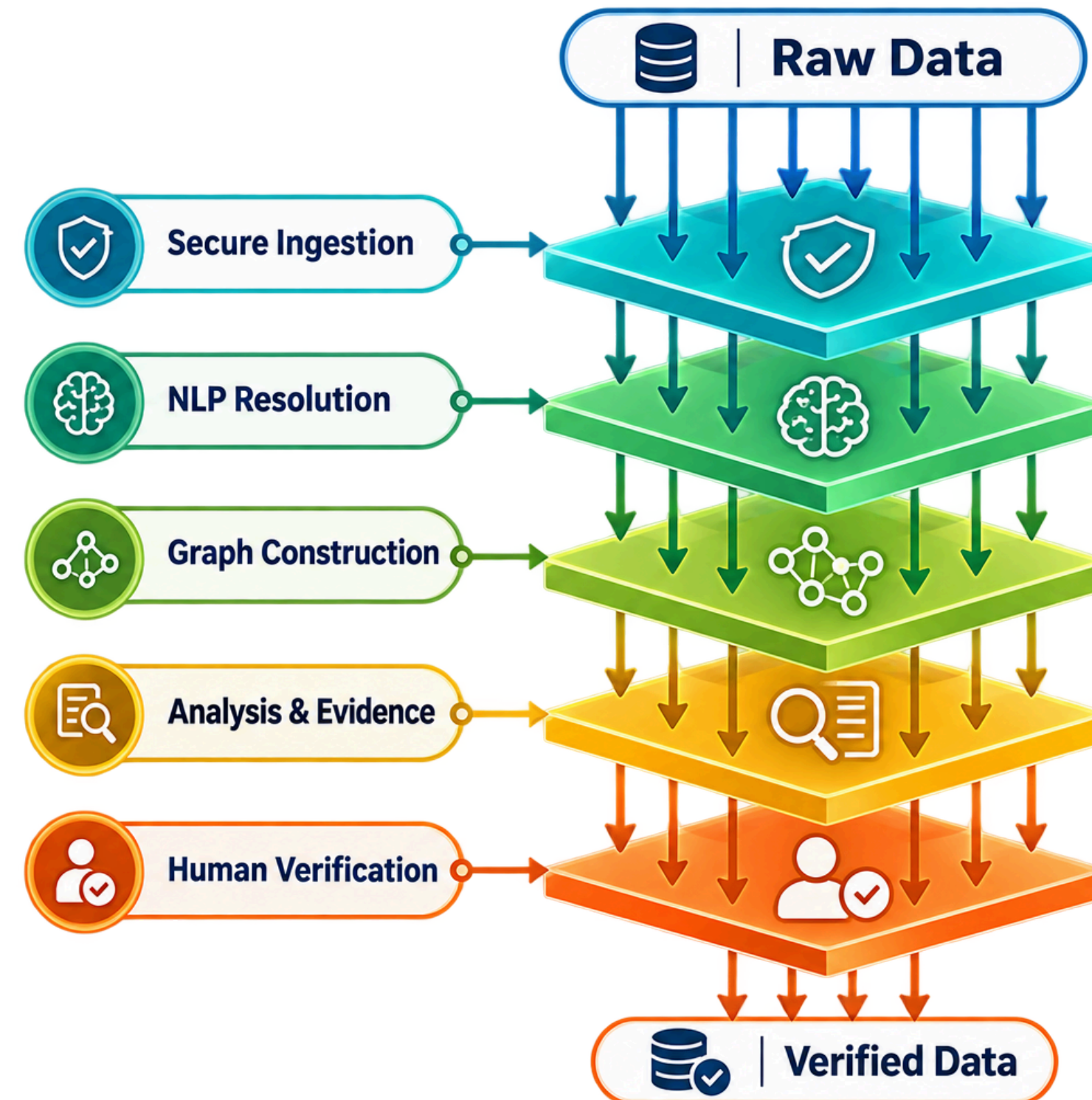
AI / ML

- NLP : AI4Bharat IndicNLP • Bhashini API
- Transliteration / NER : IndicXlit • IndicNER
- Security Model : Qwen2.5-Coder 14B + Custom LoRA
- Inference : vLLM • Sentence Transformers

Local Hardware

- GPU : NVIDIA RTX 3090 / 4090
- 24 GB VRAM — Air-gapped 14B inference

METHODOLOGY / PROCESS FLOW



FEASIBILITY —

WHY RAKSHAK AI CAN BE DEPLOYED

- 1**

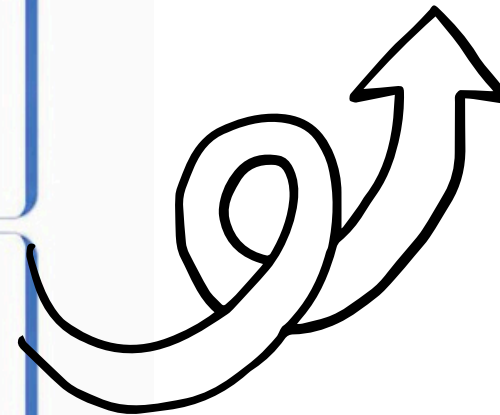
AIR-GAPPED & ON-PREMISE DEPLOYMENT
 GoI data sovereignty mandates local hosting. Our stack (vLLM + local models) runs entirely offline without cloud APIs.
- 2**

CCTNS / NCRB ECOSYSTEM READY
 Built on standard REST APIs, allowing seamless integration with existing police databases like CCTNS without altering legacy systems.
- 3**

MODULAR ROLLOUT STRATEGY
 No national rip-and-replace. Deploy individually—FIR Assistant, ChargeSheet Copilot, Case Summarizer—based on unit needs and scale progressively.
- 4**

SECURE BY DESIGN
 All data stays within the department. Role-based access, audit logs, encryption at rest & in transit ensure enterprise-grade security and compliance.
- 5**

SCALABLE & MAINTAINABLE ARCHITECTURE
 vLLM-powered inference for high performance on local GPUs. Containerized, observable, and easy to update—built for long-term sustainability.



CHALLENGES & RISKS

- 01**

HINGLISH & TRANSLITERATION NOISE
 Accurate NER and Entity Resolution for code-mixed text (e.g., “bhai ne unauthorized construction kar liya”).
- 02**

DATA QUALITY & AVAILABILITY
 Variability in municipal records, missing fields, and inconsistent formats.
- 03**

LEGAL & POLICY COMPLEXITY
 Varying bye-laws across ULBs and evolving regulatory frameworks.
- 04**

PRIVACY & SECURITY
 Handling sensitive citizen and property data with strong access controls and compliance.
- 05**

SCALABILITY & ADOPTION
 Ensuring system performance at scale and driving user adoption across stakeholders.

IMPACT — WHAT RAKSHAK AI CHANGES

- 01**

FASTER INVESTIGATION
 Reduce manual cross-searching across fragmented police records and surface relevant leads faster.
- 02**

CONNECTED INTELLIGENCE
 Connect persons, vehicles, locations and incidents across otherwise isolated cases.
- 03**

EVIDENCE DISCOVERY
 Link analytical leads back to source records with traceability and evidentiary integrity.
- 04**

DATA-DRIVEN DECISIONS
 Enable officers to act on insights, patterns and risk signals backed by trusted data.
- 05**

SECURE & COMPLIANT
 Built with role-based access, audit trails, encryption and privacy-by-design to ensure secure and lawful use of data.
- 06**

SAFER COMMUNITIES
 Faster case resolution, better resource allocation and proactive policing for greater public safety.

BENEFITS

— VALUE CREATED BY RAKSHAK AI —

- 01**
POLICE

➤ Faster Lead Discovery
 Less manual searching across FIRs, records and connected cases.
- 02**
INVESTIGATORS

➤ Better Case Correlation
 Person-vehicle-location-incident relationships reveal hidden connections.
- 03**
GOVERNMENT

➤ API-First Integration
 Connect existing police databases without requiring immediate legacy-system migration.
- 04**
PUBLIC PROSECUTORS

➤ Stronger, Evidence-Backed Cases
 AI-curated timelines, links and evidence strengthen case preparation.
- 05**
CITIZENS

➤ Safer Communities
 Faster investigations and smarter policing lead to greater public safety.
- 06**
POLICE ADMINISTRATION

➤ Higher Efficiency, Lower Cost
 Automated intelligence and streamlined workflows reduce workload and costs.

- **Graph Theory & Criminal Networks:**

- Rossi et al. (2020) - Temporal Graph Networks (TGNs) [Used for temporal edge evolution & link prediction logic]
- De Domenico et al. (2013) - Multilayer Network Analysis [Foundation for our 6-layer graph architecture]
- Ribeiro et al. - GNNs in Criminal Networks [Baseline study for community detection vs. classical algorithms]

- **Entity Resolution & Indian NLP :**

- Christen, P. (2012) - Data Matching: Concepts and Techniques [Formalization of our Hybrid ER scoring function]
- Beniwal & Singh (2023) - COMI-LINGUA Dataset [Baseline for our Hinglish code-mixed NER handling]
- AI4Bharat & Bhashini (GoI) - IndicXlit & IndicNER [Open-source tools used for Devanagari-Roman transliteration]

- **Explainability & Human-AI Teaming :**

- Ying et al. (2019) - GNNExplainer [Conceptual basis for our Evidence Panel traceability]
- Yuan et al. - Human-in-the-Loop AI Systems [Research backing our mandatory verification workflow]

- **Proprietary Infrastructure (Our Work) :**

- RakshakAI (2025) - Muneerali199/rakshak-cwe-14b-sft-final

 [\[https://huggingface.co/Muneerali199/rakshak-cwe-14b-sft-final\]](https://huggingface.co/Muneerali199/rakshak-cwe-14b-sft-final)

(Custom Qwen2.5-Coder 14B LoRA fine-tune for secure-by-design code analysis)

